Scientific Analysis of Moral Judgments using Tobit and CLAD Models

Automated Research Pipeline

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1 Dataset and Sample Description

The empirical foundation of this project rests on two primary experimental datasets: FLORIDA and BUC. The sample consists of students from the Bucaramanga Campus. These datasets capture incentivized moral judgments from distinct socio-economic contexts. Participants were presented with standardized negotiation scenarios where a negotiator's decision resulted in varying degrees of payoff for themselves, their own group, and a victim group (either ingroup or outgroup).

2 Datacard and Variable Definitions

The following table defines the primary symbols and variables used in the mathematical specifications and hypothesis tests.

Table 1: Symbols and Variable Dictionary.

Symbol	Definition
y_{ij}	Observed moral judgment of scenario j by participant i.
y^*_{ij}	Latent moral preference score (unbounded).
β_1	Regression coefficient representing the marginal effect of the predictor.
$\beta_{0.5}$	Median-regression coefficient from the CLAD robustness model.
$Q_{0.5}(y^*_{ij} \mid \mathbf{x}_{ij})$	Conditional median of the latent bounded outcome given the predictors.
IRI_i	Empathy score (Average composite of the Interpersonal Reactivity Index).
OutgroupPerp	Binary indicator: 1 if the scenario perpetrator is an outgroup member.
SameGroupHarm	Binary indicator: 1 if the harm is inflicted on the perpetrator's own group.
ICC	Intraclass Correlation: ratio of between-cluster variance to total variance.

3 Hypotheses to Test

- **H1 (Empathy):** Higher empathy predicts lower moral-judgment scores for harmful decisions.
- **H2a (Betrayal):** Same-faculty harm receives lower moral-judgment scores than cross-faculty harm.
- **H2b (Outgroup):** Outgroup perpetrators receive lower moral-judgment scores than ingroup perpetrators.

- **H3 (Moderation):** The empathy effect is stronger in outgroup cases.

4 Mathematical Approach and Theoretical Foundations

The analysis employs two complementary estimators for bounded moral judgments. The primary specification is a Two-Limit Tobit model, which is theoretically appropriate for dependent variables that are strictly bounded within a known interval. In this context, moral judgments (y_{ij}) are observed on a scale from -9 to 9. The Tobit model assumes the existence of a latent, unobserved preference index (y^*_{ij}) that follows a linear relationship:

$$y^*_{ij} = \mathbf{x}'_{ij}\beta + \epsilon_{ij}, \quad \epsilon_{ij} \sim N(0, \sigma^2)$$

The actual observed judgment y_{ij} relates to this latent variable via the censoring transformation:

$$y_{ij} = \max(-9, \min(9, y^*_{ij}))$$

This approach prevents the 'ceiling' and 'floor' effects from biasing the linear coefficients, as would occur in standard OLS regression.

Because the Tobit model relies on a Gaussian latent-error assumption, the pipeline also fits a CLAD robustness specification implemented as interval-censored median quantile regression ($p = 0.5$). This complementary estimator targets the conditional median of the latent bounded outcome and is less sensitive to heavy tails and non-normal disturbances:

$$Q_{0.5}(y^*_{ij} \mid \mathbf{x}_{ij}) = \mathbf{x}'_{ij}\beta_{0.5}$$

The CLAD branch preserves the censoring structure while relaxing the parametric normality assumption. The Tobit and CLAD results should therefore be interpreted jointly: Tobit provides the clustered parametric benchmark, while CLAD serves as a robustness check against misspecified latent-error distributions.

5 Analysis of Sample Size Impact

The statistical validity of our inferences depends on the trade-off between False Positives (Type I error, α) and False Negatives (Type II error, β). Given the clustered nature of our data, sample size impact is not merely a count of observations, but a function of cluster correlation.

5.0.1 Step-by-Step Sensitivity Analysis

To determine the impact of our sample size on our ability to detect effects, we follow these steps:

1. Calculate the Design Effect (Deff): As multiple judgments are nested within individuals, we adjust for the Intraclass Correlation (ICC).

$$Deff = 1 + (m - 1) \times ICC$$

where m is the average number of scenarios per participant.

2. Determine the Effective Sample Size (ESS): The ESS represents the number of independent observations that would provide the same statistical power as our clustered sample.

$$ESS = \frac{n_{total}}{Deff}$$

3. Inference Impact: A higher ICC reduces the ESS, thereby increasing the Standard Error of our Tobit coefficients. If the ESS is low, our models become 'conservative', increasing the risk of Type II errors (failing to support a hypothesis). By using clustered robust standard errors, we ensure that our p-values acknowledge this reduced information density, protecting the integrity of our Type I error threshold ($\alpha = 0.05$).

Based on the current run, the observed Intraclass Correlation (ICC) is 0.704, with an Effective Sample Size (ESS) of 104.8.

6 Descriptive Statistics

The empathy profile of the sample is visualized in Figure 1, showing the average scores across the four IRI latent variables.

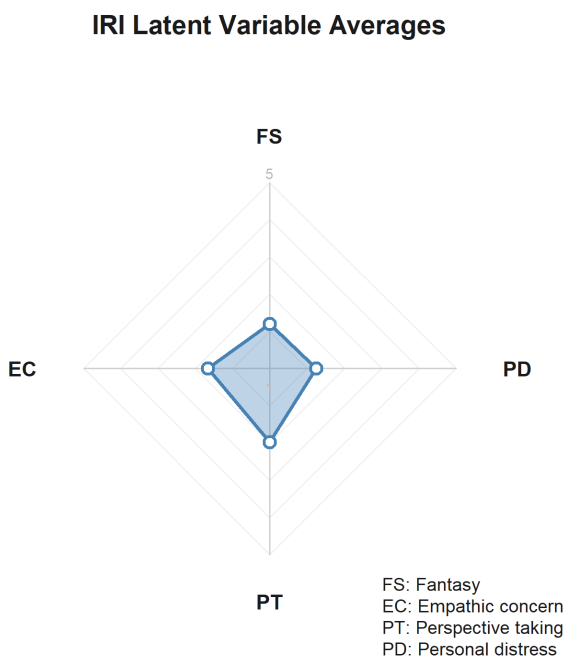

Figure 1: IRI Latent Variable Averages (Radar Plot profile).

Judgment severity distributions segmented by experimental conditions are presented in Figure 2.

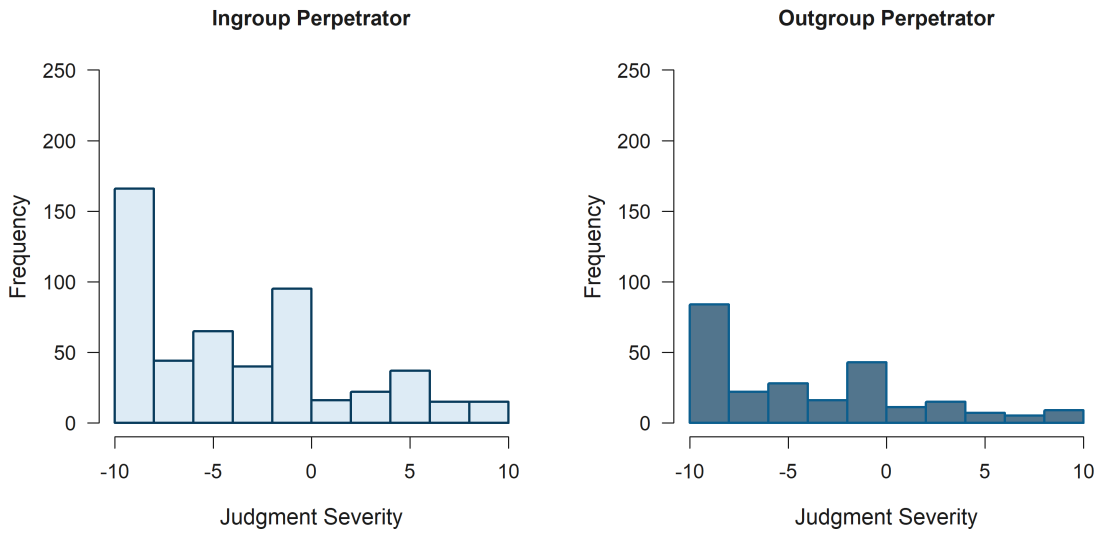

Figure 2: Judgment Severity Distribution (Ingroup vs. Outgroup).

7 Bi-variate Statistics

The correlation matrix between the psychometric subscales and the mean moral judgment is presented below.

Table 2: Correlation Matrix: IRI Subscales and Moral Judgment.

Fantasy	Empathic Concern	Perspective Taking	Personal Distress	Total IRI	Mean Judgment
1.000	0.432	0.482	0.480	0.793	0.062
0.432	1.000	0.454	0.641	0.799	-0.003
0.482	0.454	1.000	0.301	0.744	0.162
0.480	0.641	0.301	1.000	0.756	-0.009
0.793	0.799	0.744	0.756	1.000	0.075
0.062	-0.003	0.162	-0.009	0.075	1.000

Visual representations of these relationships are provided in Figure 3.

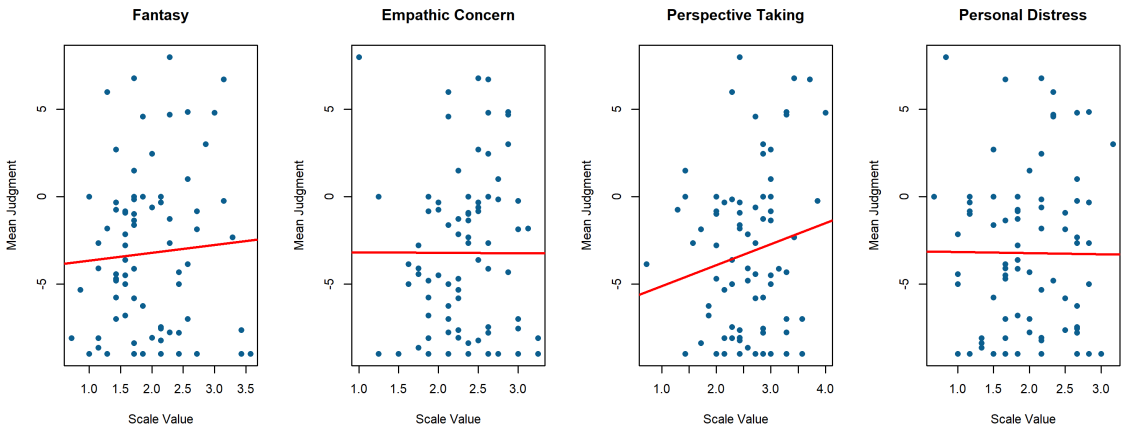

Figure 3: Bivariate Scatters: IRI Scales vs. Mean Judgment.

8 Estimator Fit Summary

The following table consolidates the fit-status information for the primary Tobit models and the added CLAD robustness checks.

Table 3: Estimator fit summary across Tobit and CLAD specifications.

Model	Approach	Status	Converged	Iterations	Observations	Participants	LowerBoundCensored	UpperBoundCensored
H1_A_Total_CLAD	CLAD	not_converged	FALSE	2000	755	77	227	24
H1_B_Constructs_CLAD	CLAD	completed	TRUE	109	755	77	227	24
H2a_A_Total_CLAD	CLAD	completed	TRUE	23	503	77	160	18
H2a_B_Constructs_CLAD	CLAD	not_converged	FALSE	2000	503	77	160	18
H2b_A_Total_CLAD	CLAD	not_converged	FALSE	2000	755	77	227	24
H2b_B_Constructs_CLAD	CLAD	completed	TRUE	108	755	77	227	24
H3_A_Total_CLAD	CLAD	not_converged	FALSE	2000	755	77	227	24
H3_B_Constructs_CLAD	CLAD	completed	TRUE	108	755	77	227	24
H1_A_Total_Tobit	Tobit	completed	NA	3	755	77	227	24
H1_B_Constructs_Tobit	Tobit	completed	NA	3	755	77	227	24
H2a_A_Total_Tobit	Tobit	completed	NA	3	503	77	160	18
H2a_B_Constructs_Tobit	Tobit	completed	NA	3	503	77	160	18
H2b_A_Total_Tobit	Tobit	completed	NA	3	755	77	227	24
H2b_B_Constructs_Tobit	Tobit	completed	NA	3	755	77	227	24
H3_A_Total_Tobit	Tobit	completed	NA	3	755	77	227	24
H3_B_Constructs_Tobit	Tobit	completed	NA	3	755	77	227	24

8.1 Integrated Hypothesis Conclusions

The following summary restates each original hypothesis and indicates whether the available Tobit estimates and the converged CLAD robustness models support it in the current data.

- H1. Original hypothesis: Higher empathy predicts lower moral-judgment scores for harmful decisions. Tobit conclusion: the available models do not support the hypothesis. Model A does not support the hypothesis; Empathy composite (average) is negative but not statistically significant ($p = 0.475$). Model B does not support the hypothesis; none of the empathy subscale main effects are statistically significant, and the closest signal is Empathy: Perspective taking with a positive association ($p = 0.159$). Additional statistically significant signals include Personal distress x outgroup perpetrator with a positive association ($p = 0.013$) and Age with a positive association ($p = 0.036$). CLAD conclusion: the available models do not support the hypothesis. Only converged CLAD specifications are interpreted here. Model A is

not interpreted because the CLAD optimization did not converge after 2000 iterations. Model B does not support the hypothesis; none of the empathy subscale main effects are statistically significant, and the closest signal is Empathy: Personal distress with a negative association ($p = 0.680$).

- H2a. Original hypothesis: Same-faculty harm receives lower moral-judgment scores than cross-faculty harm. Tobit conclusion: the available models do not support the hypothesis. Model A does not support the hypothesis; Negotiator and victim share faculty is positive but not statistically significant ($p = 0.920$). Model B does not support the hypothesis; Negotiator and victim share faculty is positive but not statistically significant ($p = 0.879$). Additional statistically significant signals include Age with a positive association ($p = 0.015$). CLAD conclusion: the available models do not support the hypothesis. Only converged CLAD specifications are interpreted here. Model A does not support the hypothesis; Negotiator and victim share faculty is positive but not statistically significant ($p = 0.874$). Model B is not interpreted because the CLAD optimization did not converge after 2000 iterations. Additional statistically significant signals include Man (ref = woman) with a negative association ($p = 0.002$) and Empathy composite (average) with a negative association ($p = 0.003$).

- H2b. Original hypothesis: Outgroup perpetrators receive lower moral-judgment scores than ingroup perpetrators. Tobit conclusion: the available models do not support the hypothesis. Model A does not support the hypothesis; Outgroup perpetrator (ref = ingroup) is negative but not statistically significant ($p = 0.446$). Model B does not support the hypothesis; Outgroup perpetrator (ref = ingroup) is negative but not statistically significant ($p = 0.813$). Additional statistically significant signals include Personal distress x outgroup perpetrator with a positive association ($p = 0.013$) and Age with a positive association ($p = 0.036$). CLAD conclusion: the available models do not support the hypothesis. Only converged CLAD specifications are interpreted here. Model A is not interpreted because the CLAD optimization did not converge after 2000 iterations. Model B does not support the hypothesis; Outgroup perpetrator (ref = ingroup) is negative but not statistically significant ($p = 0.884$).

- H3. Original hypothesis: The empathy effect is stronger in outgroup cases than in ingroup cases. Tobit conclusion: the available models do not support the hypothesis and at least one model points in the opposite direction. Model A does not support the hypothesis; Empathy x outgroup perpetrator is positive but not statistically significant ($p = 0.477$). Model B points in the opposite direction through Personal distress x outgroup perpetrator with a positive association ($p = 0.013$). Additional statistically significant signals include Age with a positive association ($p = 0.036$). CLAD conclusion: the available models do not support the hypothesis. Only converged CLAD specifications are interpreted here. Model A is not interpreted because the CLAD optimization did not converge after 2000 iterations. Model B does not support the hypothesis; none of the empathy-dimension x outgroup interactions are statistically significant, and the closest signal is Personal distress x outgroup perpetrator with a positive association ($p = 0.494$).

9 Hypothesis Validation and Results

Detailed coefficient tables for each Tobit model, coupled with CLAD robustness replications, natural language interpretive narratives, and estimator-specific diagnostics, are provided below.

9.1 H1: Empathy Effect

This section evaluates the primary effect of empathy on moral judgments.

9.1.1 Model A: Composite Empathy

Table 4: H1 Model A: Composite Empathy Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-9.592	6.858	0.162
Empathy composite (average)	-1.668	2.337	0.475
Outgroup perpetrator (ref = ingroup)	-2.077	2.725	0.446
Control label hidden (ref = ingroup)	-2.658	3.346	0.427
Victim outgroup (ref = ingroup)	0.278	0.910	0.760
Observer role (ref = victim)	0.545	0.798	0.494
Engineering participant (ref = humanities)	0.188	1.556	0.904
Man (ref = woman)	-1.765	1.744	0.312
Age	0.478	0.241	0.047
Socioeconomic status	-0.247	0.706	0.726
Negotiator 2 (ref = negotiator 1)	0.124	0.458	0.787
Empathy x outgroup perpetrator	0.885	1.245	0.477
Empathy x control label hidden	1.578	1.589	0.321
Scale variance (log)	1.993	0.096	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -9.592 (p=0.162). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -1.668. The p-value of 0.475 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -2.077. The p-value of 0.446 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of -2.658. The p-value of 0.427 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.278. The p-value of 0.760 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.545. The p-value of 0.494 indicates the effect is not statistically significant, meaning we do not have enough

evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.188. The p-value of 0.904 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -1.765. The p-value of 0.312 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.478. The p-value of 0.047 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.247. The p-value of 0.726 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:perp_outgroup (representing the combined interaction effect between overall empathy and an outgroup perpetrator) has an estimated β coefficient of 0.885. The p-value of 0.477 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:perp_control (representing the combined interaction effect between overall empathy and an unidentified perpetrator) has an estimated β coefficient of 1.578. The p-value of 0.321 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.993, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.934, p = 8.055e-18). Consequently, standard clustered errors might be inefficient. For maximum robustness in future analyses, we recommend exploring a non-parametric alternative such as Censored Least Absolute Deviations (CLAD) or specifying bootstrapped standard errors for the Tobit estimator.

Table 5: H1 Model A: Composite Empathy Regression Coefficients (CLAD robustness).

label	estimate	std_error	p_value
Intercept	-12.967	7.538	0.085
Empathy composite (average)	-1.611	1.551	0.299
Outgroup perpetrator (ref = ingroup)	-1.443	3.948	0.715
Control label hidden (ref = ingroup)	-4.722	5.435	0.385
Victim outgroup (ref = ingroup)	-0.328	0.532	0.537
Observer role (ref = victim)	1.093	0.669	0.102
Engineering participant (ref = humanities)	0.066	0.496	0.894
Man (ref = woman)	-0.803	0.706	0.256
Age	0.667	0.286	0.020
Socioeconomic status	-0.551	0.304	0.070
Negotiator 2 (ref = negotiator 1)	0.093	0.429	0.828
Empathy x outgroup perpetrator	0.624	1.891	0.741
Empathy x control label hidden	2.443	2.704	0.366

CLAD Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -12.967 (p=0.085). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -1.611. The p-value of 0.299 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -1.443. The p-value of 0.715 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of -4.722. The p-value of 0.385 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of -0.328. The p-value of 0.537 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 1.093. The p-value of 0.102 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.066. The p-value of 0.894 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.803. The p-value of 0.256 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's

biological age in years) has an estimated β coefficient of 0.667. The p-value of 0.020 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.551. The p-value of 0.070 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:perp_outgroup (representing the combined interaction effect between overall empathy and an outgroup perpetrator) has an estimated β coefficient of 0.624. The p-value of 0.741 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:perp_control (representing the combined interaction effect between overall empathy and an unidentified perpetrator) has an estimated β coefficient of 2.443. The p-value of 0.366 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The CLAD robustness model was estimated as an interval-censored median regression (p = 0.50). This estimator does not impose a Gaussian latent-error assumption and is therefore reported as a robustness check alongside the Tobit model when normality is doubtful. The optimization status for this fit is not confirmed as converged after 2000 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units; the reported coefficients are back-transformed to the original judgement scale. Standard errors come from the ctqr asymptotic covariance matrix and should be interpreted as complementary robustness evidence rather than clustered Tobit replacements.

9.1.2 Model B: Separated Empathy Constructs

Table 6: H1 Model B: Separated Constructs Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-11.669	7.497	0.120
Empathy: Fantasy scale	-1.347	1.766	0.446
Empathy: Empathic concern	-1.294	2.639	0.624
Empathy: Perspective taking	2.558	1.815	0.159
Empathy: Personal distress	-1.397	1.866	0.454
Outgroup perpetrator (ref = ingroup)	-0.751	3.182	0.813
Control label hidden (ref = ingroup)	-1.828	4.018	0.649
Victim outgroup (ref = ingroup)	0.180	0.880	0.838
Observer role (ref = victim)	0.425	0.783	0.587
Engineering participant (ref = humanities)	0.530	1.660	0.749
Man (ref = woman)	-1.794	1.825	0.325
Age	0.489	0.233	0.036
Socioeconomic status	-0.300	0.688	0.663
Negotiator 2 (ref = negotiator 1)	0.140	0.452	0.757
Fantasy x outgroup perpetrator	-0.571	1.157	0.622
Empathic concern x outgroup perpetrator	-0.959	1.289	0.457
Perspective taking x outgroup perpetrator	-0.953	1.145	0.405
Personal distress x outgroup perpetrator	3.287	1.317	0.013
Fantasy x control label hidden	-0.107	1.477	0.942
Empathic concern x control label hidden	0.102	1.568	0.948
Perspective taking x control label hidden	0.357	1.836	0.846
Personal distress x control label hidden	0.840	1.247	0.501
Scale variance (log)	1.976	0.093	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -11.669 (p=0.120). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -1.347. The p-value of 0.446 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -1.294. The p-value of 0.624 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an esti-

mated β coefficient of 2.558. The p-value of 0.159 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -1.397. The p-value of 0.454 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.751. The p-value of 0.813 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of -1.828. The p-value of 0.649 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.180. The p-value of 0.838 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.425. The p-value of 0.587 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.530. The p-value of 0.749 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -1.794. The p-value of 0.325 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.489. The p-value of 0.036 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.300. The p-value of 0.663 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:perp_outgroup (representing the combined interaction effect between the Fantasy scale and an outgroup perpetrator) has an estimated β coefficient of -0.571. The p-value of 0.622 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_outgroup (representing the combined interaction effect between Empathic Concern and an outgroup perpetrator) has an estimated β coefficient of -0.959. The p-value of 0.457 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:perp_outgroup (representing the combined interaction effect between Perspective Taking and an outgroup perpetrator) has an estimated β coefficient of -0.953. The p-value of 0.405 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_outgroup (representing the combined interaction effect between Personal Distress and an outgroup perpetrator) has an estimated β coefficient of 3.287. The p-value of 0.013 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_fs:perp_control (representing the combined interaction effect between the Fantasy scale and an unidentified perpetrator) has an estimated β coefficient of -0.107. The p-value of 0.942 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_control (representing

the combined interaction effect between Empathic Concern and an unidentified perpetrator) has an estimated β coefficient of 0.102. The p-value of 0.948 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:perp_control (representing the combined interaction effect between Perspective Taking and an unidentified perpetrator) has an estimated β coefficient of 0.357. The p-value of 0.846 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_control (representing the combined interaction effect between Personal Distress and an unidentified perpetrator) has an estimated β coefficient of 0.840. The p-value of 0.501 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.976, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.951, p = 3.347e-15). Consequently, standard clustered errors might be inefficient. For maximum robustness in future analyses, we recommend exploring a non-parametric alternative such as Censored Least Absolute Deviations (CLAD) or specifying bootstrapped standard errors for the Tobit estimator.

Table 7: H1 Model B: Separated Constructs Regression Coefficients (CLAD robustness).

label	estimate	std_error	p_value
Intercept	-10.262	19.150	0.592
Empathy: Fantasy scale	-0.364	12.100	0.976
Empathy: Empathic concern	-0.557	12.911	0.966
Empathy: Perspective taking	0.920	3.652	0.801
Empathy: Personal distress	-1.928	4.670	0.680
Outgroup perpetrator (ref = ingroup)	-2.753	18.819	0.884
Control label hidden (ref = ingroup)	-6.332	24.214	0.794
Victim outgroup (ref = ingroup)	-0.630	1.079	0.559
Observer role (ref = victim)	0.974	0.699	0.164
Engineering participant (ref = humanities)	0.136	0.651	0.835
Man (ref = woman)	-0.961	3.102	0.757
Age	0.528	0.488	0.280
Socioeconomic status	-0.551	0.572	0.335
Negotiator 2 (ref = negotiator 1)	-0.052	0.518	0.919
Fantasy x outgroup perpetrator	-0.593	11.581	0.959
Empathic concern x outgroup perpetrator	0.385	11.557	0.973
Perspective taking x outgroup perpetrator	-0.941	5.443	0.863
Personal distress x outgroup perpetrator	2.562	3.742	0.494
Fantasy x control label hidden	-0.569	11.930	0.962
Empathic concern x control label hidden	0.961	15.695	0.951
Perspective taking x control label hidden	1.385	4.819	0.774
Personal distress x control label hidden	1.052	4.292	0.806

CLAD Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -10.262 (p=0.592). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.364. The p-value of 0.976 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -0.557. The p-value of 0.966 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.920. The p-value of 0.801 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of

-1.928. The p-value of 0.680 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -2.753. The p-value of 0.884 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of -6.332. The p-value of 0.794 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of -0.630. The p-value of 0.559 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.974. The p-value of 0.164 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.136. The p-value of 0.835 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.961. The p-value of 0.757 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.528. The p-value of 0.280 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.551. The p-value of 0.335 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:perp_outgroup (representing the combined interaction effect between the Fantasy scale and an outgroup perpetrator) has an estimated β coefficient of -0.593. The p-value of 0.959 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_outgroup (representing the combined interaction effect between Empathic Concern and an outgroup perpetrator) has an estimated β coefficient of 0.385. The p-value of 0.973 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:perp_outgroup (representing the combined interaction effect between Perspective Taking and an outgroup perpetrator) has an estimated β coefficient of -0.941. The p-value of 0.863 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_outgroup (representing the combined interaction effect between Personal Distress and an outgroup perpetrator) has an estimated β coefficient of 2.562. The p-value of 0.494 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:perp_control (representing the combined interaction effect between the Fantasy scale and an unidentified perpetrator) has an estimated β coefficient of -0.569. The p-value of 0.962 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_control (representing the combined interaction effect between Empathic Concern and an unidentified perpetrator) has an estimated β coefficient of 0.961. The p-value of 0.951 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:perp_control

(representing the combined interaction effect between Perspective Taking and an unidentified perpetrator) has an estimated β coefficient of 1.385. The p-value of 0.774 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_control (representing the combined interaction effect between Personal Distress and an unidentified perpetrator) has an estimated β coefficient of 1.052. The p-value of 0.806 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The CLAD robustness model was estimated as an interval-censored median regression (p = 0.50). This estimator does not impose a Gaussian latent-error assumption and is therefore reported as a robustness check alongside the Tobit model when normality is doubtful. The optimization status for this fit is converged after 109 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units; the reported coefficients are back-transformed to the original judgement scale. Standard errors come from the ctqr asymptotic covariance matrix and should be interpreted as complementary robustness evidence rather than clustered Tobit replacements.

9.2 H2a: Ingroup Betrayal

This section tests whether harm directed at the perpetrator's own group is judged differently.

9.2.1 Model A: Composite Empathy Control

Table 8: H2a Model A: Composite Empathy Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-11.054	6.721	0.100
Negotiator and victim share faculty	0.081	0.802	0.920
Empathy composite (average)	-1.552	2.327	0.505
Outgroup perpetrator (ref = ingroup)	-0.004	0.707	0.995
Victim outgroup (ref = ingroup)	0.465	1.236	0.707
Observer role (ref = victim)	0.564	0.840	0.502
Engineering participant (ref = humanities)	-0.526	1.602	0.743
Man (ref = woman)	-2.123	1.867	0.256
Age	0.541	0.227	0.017
Socioeconomic status	-0.070	0.722	0.923
Negotiator 2 (ref = negotiator 1)	0.147	0.597	0.806
Scale variance (log)	2.006	0.101	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -11.054 (p=0.100). The term same_group_harm (representing whether the harm inflicted by the perpetrator targeted a victim from their own faculty (Ingroup Betrayal)) has an estimated β

coefficient of 0.081. The p-value of 0.920 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -1.552. The p-value of 0.505 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.004. The p-value of 0.995 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.465. The p-value of 0.707 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.564. The p-value of 0.502 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.526. The p-value of 0.743 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -2.123. The p-value of 0.256 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.541. The p-value of 0.017 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.070. The p-value of 0.923 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 2.006, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.932, p = 2.246e-14). Consequently, standard clustered errors might be inefficient. For maximum robustness in future analyses, we recommend exploring a non-parametric alternative such as Censored Least Absolute Deviations (CLAD) or specifying bootstrapped standard errors for the Tobit estimator.

Table 9: H2a Model A: Composite Empathy Regression Coefficients (CLAD robustness).

label	estimate	std_error	p_value
Intercept	-13.291	10.498	0.205
Negotiator and victim share faculty	0.100	0.633	0.874
Empathy composite (average)	-2.028	0.675	0.003
Outgroup perpetrator (ref = ingroup)	-0.034	0.496	0.946
Victim outgroup (ref = ingroup)	-0.017	0.841	0.984
Observer role (ref = victim)	0.916	0.855	0.284
Engineering participant (ref = humanities)	-0.225	0.732	0.759
Man (ref = woman)	-1.489	0.471	0.002
Age	0.722	0.484	0.135
Socioeconomic status	-0.423	0.368	0.251
Negotiator 2 (ref = negotiator 1)	0.257	0.524	0.624

CLAD Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -13.291 (p=0.205). The term same_group_harm (representing whether the harm inflicted by the perpetrator targeted a victim from their own faculty (Ingroup Betrayal)) has an estimated β coefficient of 0.100. The p-value of 0.874 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -2.028. The p-value of 0.003 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.034. The p-value of 0.946 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of -0.017. The p-value of 0.984 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.916. The p-value of 0.284 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.225. The p-value of 0.759 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -1.489. The p-value of 0.002 indicates statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.722. The p-value of 0.135 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status

(representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.423. The p-value of 0.251 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The CLAD robustness model was estimated as an interval-censored median regression (p = 0.50). This estimator does not impose a Gaussian latent-error assumption and is therefore reported as a robustness check alongside the Tobit model when normality is doubtful. The optimization status for this fit is converged after 23 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units; the reported coefficients are back-transformed to the original judgement scale. Standard errors come from the ctqr asymptotic covariance matrix and should be interpreted as complementary robustness evidence rather than clustered Tobit replacements.

9.2.2 Model B: Separated Construct Controls

Table 10: H2a Model B: Separated Constructs Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-12.446	7.706	0.106
Negotiator and victim share faculty	0.125	0.819	0.879
Empathy: Fantasy scale	-1.818	1.800	0.312
Empathy: Empathic concern	-1.449	2.800	0.605
Empathy: Perspective taking	1.893	1.716	0.270
Empathy: Personal distress	-0.019	1.707	0.991
Outgroup perpetrator (ref = ingroup)	0.129	0.678	0.849
Victim outgroup (ref = ingroup)	0.501	1.211	0.679
Observer role (ref = victim)	0.425	0.827	0.607
Engineering participant (ref = humanities)	-0.197	1.746	0.910
Man (ref = woman)	-2.103	2.028	0.300
Age	0.547	0.225	0.015
Socioeconomic status	-0.186	0.729	0.798
Negotiator 2 (ref = negotiator 1)	0.053	0.564	0.925
Scale variance (log)	1.998	0.099	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -12.446 (p=0.106). The term same_group_harm (representing whether the harm inflicted by the perpetrator targeted a victim from their own faculty (Ingroup Betrayal)) has an estimated β coefficient of 0.125. The p-value of 0.879 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -1.818. The p-value of 0.312

indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -1.449. The p-value of 0.605 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 1.893. The p-value of 0.270 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -0.019. The p-value of 0.991 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of 0.129. The p-value of 0.849 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.501. The p-value of 0.679 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.425. The p-value of 0.607 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.197. The p-value of 0.910 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -2.103. The p-value of 0.300 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.547. The p-value of 0.015 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.186. The p-value of 0.798 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.998, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.943, p = 6.339e-13). Consequently, standard clustered errors might be inefficient. For maximum robustness in future analyses, we recommend exploring a non-parametric alternative such as Censored Least Absolute Deviations (CLAD) or specifying bootstrapped standard errors for the Tobit estimator.

Table 11: H2a Model B: Separated Constructs Regression Coefficients (CLAD robustness).

label	estimate	std_error	p_value
Intercept	-13.941	17.450	0.424
Negotiator and victim share faculty	0.141	0.376	0.709
Empathy: Fantasy scale	-1.034	1.820	0.570
Empathy: Empathic concern	-0.075	2.417	0.975
Empathy: Perspective taking	0.384	0.508	0.450
Empathy: Personal distress	-1.194	1.661	0.472
Outgroup perpetrator (ref = ingroup)	0.087	0.458	0.849
Victim outgroup (ref = ingroup)	0.087	1.141	0.939
Observer role (ref = victim)	0.436	0.978	0.656
Engineering participant (ref = humanities)	-0.087	0.917	0.924
Man (ref = woman)	-1.903	0.481	0.000
Age	0.732	0.787	0.352
Socioeconomic status	-0.478	0.580	0.410
Negotiator 2 (ref = negotiator 1)	0.053	0.422	0.899

CLAD Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -13.941 (p=0.424). The term same_group_harm (representing whether the harm inflicted by the perpetrator targeted a victim from their own faculty (Ingroup Betrayal)) has an estimated β coefficient of 0.141. The p-value of 0.709 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -1.034. The p-value of 0.570 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -0.075. The p-value of 0.975 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.384. The p-value of 0.450 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -1.194. The p-value of 0.472 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of 0.087. The p-value of 0.849 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the

victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.087. The p-value of 0.939 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.436. The p-value of 0.656 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of -0.087. The p-value of 0.924 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -1.903. The p-value of 0.000 indicates highly statistically significant, meaning the predictor has a negative effect on the latent moral judgment. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.732. The p-value of 0.352 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.478. The p-value of 0.410 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The CLAD robustness model was estimated as an interval-censored median regression (p = 0.50). This estimator does not impose a Gaussian latent-error assumption and is therefore reported as a robustness check alongside the Tobit model when normality is doubtful. The optimization status for this fit is not confirmed as converged after 2000 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units; the reported coefficients are back-transformed to the original judgement scale. Standard errors come from the ctqr asymptotic covariance matrix and should be interpreted as complementary robustness evidence rather than clustered Tobit replacements.

9.3 H2b: Outgroup Derogation

This section examines the difference in judgment when the perpetrator is an outgroup member compared to an ingroup member.

9.3.1 Model A: Composite Empathy Control

Table 12: H2b Model A: Composite Empathy Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-9.592	6.858	0.162
Outgroup perpetrator (ref = ingroup)	-2.077	2.725	0.446
Empathy composite (average)	-1.668	2.337	0.475
Control label hidden (ref = ingroup)	-2.658	3.346	0.427
Victim outgroup (ref = ingroup)	0.278	0.910	0.760
Observer role (ref = victim)	0.545	0.798	0.494
Engineering participant (ref = humanities)	0.188	1.556	0.904
Man (ref = woman)	-1.765	1.744	0.312
Age	0.478	0.241	0.047
Socioeconomic status	-0.247	0.706	0.726
Negotiator 2 (ref = negotiator 1)	0.124	0.458	0.787
Empathy x outgroup perpetrator	0.885	1.245	0.477
Empathy x control label hidden	1.578	1.589	0.321
Scale variance (log)	1.993	0.096	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -9.592 (p=0.162). The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -2.077. The p-value of 0.446 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -1.668. The p-value of 0.475 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of -2.658. The p-value of 0.427 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.278. The p-value of 0.760 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.545. The p-value of 0.494 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.188. The p-value of 0.904 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -1.765. The p-value of 0.312 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.478. The p-value of 0.047 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.247. The p-value of 0.726 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_total (representing the combined interaction effect between overall empathy and an outgroup perpetrator) has an estimated β coefficient of 0.885. The p-value of 0.477 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:perp_control (representing the combined interaction effect between overall empathy and an unidentified perpetrator) has an estimated β coefficient of 1.578. The p-value of 0.321 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.993, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.934, p = 8.055e-18). Consequently, standard clustered errors might be inefficient. For maximum robustness in future analyses, we recommend exploring a non-parametric alternative such as Censored Least Absolute Deviations (CLAD) or specifying bootstrapped standard errors for the Tobit estimator.

Table 13: H2b Model A: Composite Empathy Regression Coefficients (CLAD robustness).

label	estimate	std_error	p_value
Intercept	-12.967	7.538	0.085
Outgroup perpetrator (ref = ingroup)	-1.443	3.948	0.715
Empathy composite (average)	-1.611	1.551	0.299
Control label hidden (ref = ingroup)	-4.722	5.435	0.385
Victim outgroup (ref = ingroup)	-0.328	0.532	0.537
Observer role (ref = victim)	1.093	0.669	0.102
Engineering participant (ref = humanities)	0.066	0.496	0.894
Man (ref = woman)	-0.803	0.706	0.256
Age	0.667	0.286	0.020
Socioeconomic status	-0.551	0.304	0.070
Negotiator 2 (ref = negotiator 1)	0.093	0.429	0.828
Empathy x outgroup perpetrator	0.624	1.891	0.741
Empathy x control label hidden	2.443	2.704	0.366

CLAD Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -12.967 (p=0.085). The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -1.443. The p-value of 0.715 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -1.611. The p-value of 0.299 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of -4.722. The p-value of 0.385 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of -0.328. The p-value of 0.537 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 1.093. The p-value of 0.102 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.066. The p-value of 0.894 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.803. The p-value of 0.256 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.667. The p-value of 0.020 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.551. The p-value of 0.070 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_total (representing the combined interaction effect between overall empathy and an outgroup perpetrator) has an estimated β coefficient of 0.624. The p-value of 0.741 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:perp_control (representing the combined interaction effect between overall empathy and an unidentified perpetrator) has an estimated β coefficient of 2.443. The p-value of 0.366 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The CLAD robustness model was estimated as an interval-censored median regression (p = 0.50). This estimator does not impose a Gaussian latent-error assumption and is therefore reported as a robustness check alongside the Tobit model when normality is doubtful. The optimization status for this fit is not confirmed as converged after 2000 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units; the reported coefficients are back-transformed to the original judgement scale. Standard errors come from the ctqr asymptotic covariance matrix and should be interpreted as complementary robustness evidence rather than clustered Tobit replacements.

9.3.2 Model B: Separated Construct Controls

Table 14: H2b Model B: Separated Constructs Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-11.669	7.497	0.120
Outgroup perpetrator (ref = ingroup)	-0.751	3.182	0.813
Empathy: Fantasy scale	-1.347	1.766	0.446
Empathy: Empathic concern	-1.294	2.639	0.624
Empathy: Perspective taking	2.558	1.815	0.159
Empathy: Personal distress	-1.397	1.866	0.454
Control label hidden (ref = ingroup)	-1.828	4.018	0.649
Victim outgroup (ref = ingroup)	0.180	0.880	0.838
Observer role (ref = victim)	0.425	0.783	0.587
Engineering participant (ref = humanities)	0.530	1.660	0.749
Man (ref = woman)	-1.794	1.825	0.325
Age	0.489	0.233	0.036
Socioeconomic status	-0.300	0.688	0.663
Negotiator 2 (ref = negotiator 1)	0.140	0.452	0.757
Fantasy x outgroup perpetrator	-0.571	1.157	0.622
Empathic concern x outgroup perpetrator	-0.959	1.289	0.457
Perspective taking x outgroup perpetrator	-0.953	1.145	0.405
Personal distress x outgroup perpetrator	3.287	1.317	0.013
Fantasy x control label hidden	-0.107	1.477	0.942
Empathic concern x control label hidden	0.102	1.568	0.948
Perspective taking x control label hidden	0.357	1.836	0.846
Personal distress x control label hidden	0.840	1.247	0.501
Scale variance (log)	1.976	0.093	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -11.669 (p=0.120). The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.751. The p-value of 0.813 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -1.347. The p-value of 0.446 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient

of -1.294. The p-value of 0.624 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 2.558. The p-value of 0.159 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -1.397. The p-value of 0.454 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of -1.828. The p-value of 0.649 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.180. The p-value of 0.838 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.425. The p-value of 0.587 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.530. The p-value of 0.749 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -1.794. The p-value of 0.325 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.489. The p-value of 0.036 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.300. The p-value of 0.663 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_fs (representing the combined interaction effect between the Fantasy scale and an outgroup perpetrator) has an estimated β coefficient of -0.571. The p-value of 0.622 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_ec (representing the combined interaction effect between Empathic Concern and an outgroup perpetrator) has an estimated β coefficient of -0.959. The p-value of 0.457 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_pt (representing the combined interaction effect between Perspective Taking and an outgroup perpetrator) has an estimated β coefficient of -0.953. The p-value of 0.405 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_pd (representing the combined interaction effect between Personal Distress and an outgroup perpetrator) has an estimated β coefficient of 3.287. The p-value of 0.013 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_fs:perp_control (representing the combined interaction effect between the Fantasy scale and an unidentified perpetrator) has an estimated β coefficient of -0.107. The p-value of 0.942 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_control (representing

the combined interaction effect between Empathic Concern and an unidentified perpetrator) has an estimated β coefficient of 0.102. The p-value of 0.948 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:perp_control (representing the combined interaction effect between Perspective Taking and an unidentified perpetrator) has an estimated β coefficient of 0.357. The p-value of 0.846 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_control (representing the combined interaction effect between Personal Distress and an unidentified perpetrator) has an estimated β coefficient of 0.840. The p-value of 0.501 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.976, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.951, p = 3.347e-15). Consequently, standard clustered errors might be inefficient. For maximum robustness in future analyses, we recommend exploring a non-parametric alternative such as Censored Least Absolute Deviations (CLAD) or specifying bootstrapped standard errors for the Tobit estimator.

Table 15: H2b Model B: Separated Constructs Regression Coefficients (CLAD robustness).

label	estimate	std_error	p_value
Intercept	-10.262	19.150	0.592
Outgroup perpetrator (ref = ingroup)	-2.753	18.819	0.884
Empathy: Fantasy scale	-0.364	12.100	0.976
Empathy: Empathic concern	-0.558	12.911	0.966
Empathy: Perspective taking	0.920	3.652	0.801
Empathy: Personal distress	-1.928	4.670	0.680
Control label hidden (ref = ingroup)	-6.331	24.214	0.794
Victim outgroup (ref = ingroup)	-0.630	1.079	0.559
Observer role (ref = victim)	0.974	0.699	0.164
Engineering participant (ref = humanities)	0.136	0.651	0.835
Man (ref = woman)	-0.961	3.102	0.757
Age	0.528	0.488	0.280
Socioeconomic status	-0.551	0.572	0.335
Negotiator 2 (ref = negotiator 1)	-0.052	0.518	0.919
Fantasy x outgroup perpetrator	-0.593	11.581	0.959
Empathic concern x outgroup perpetrator	0.385	11.557	0.973
Perspective taking x outgroup perpetrator	-0.941	5.443	0.863
Personal distress x outgroup perpetrator	2.562	3.742	0.494
Fantasy x control label hidden	-0.569	11.930	0.962
Empathic concern x control label hidden	0.961	15.695	0.951
Perspective taking x control label hidden	1.385	4.819	0.774
Personal distress x control label hidden	1.052	4.292	0.806

CLAD Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -10.262 (p=0.592). The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -2.753. The p-value of 0.884 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.364. The p-value of 0.976 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -0.558. The p-value of 0.966 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.920. The p-value of

0.801 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -1.928. The p-value of 0.680 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of -6.331. The p-value of 0.794 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of -0.630. The p-value of 0.559 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.974. The p-value of 0.164 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.136. The p-value of 0.835 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.961. The p-value of 0.757 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.528. The p-value of 0.280 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.551. The p-value of 0.335 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_fs (representing the combined interaction effect between the Fantasy scale and an outgroup perpetrator) has an estimated β coefficient of -0.593. The p-value of 0.959 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_ec (representing the combined interaction effect between Empathic Concern and an outgroup perpetrator) has an estimated β coefficient of 0.385. The p-value of 0.973 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_pt (representing the combined interaction effect between Perspective Taking and an outgroup perpetrator) has an estimated β coefficient of -0.941. The p-value of 0.863 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_pd (representing the combined interaction effect between Personal Distress and an outgroup perpetrator) has an estimated β coefficient of 2.562. The p-value of 0.494 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:perp_control (representing the combined interaction effect between the Fantasy scale and an unidentified perpetrator) has an estimated β coefficient of -0.569. The p-value of 0.962 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_control (representing the combined interaction effect between Empathic Concern and an unidentified perpetrator) has an estimated β coefficient of 0.961. The p-value of 0.951 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:perp_control

(representing the combined interaction effect between Perspective Taking and an unidentified perpetrator) has an estimated β coefficient of 1.385. The p-value of 0.774 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_control (representing the combined interaction effect between Personal Distress and an unidentified perpetrator) has an estimated β coefficient of 1.052. The p-value of 0.806 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The CLAD robustness model was estimated as an interval-censored median regression (p = 0.50). This estimator does not impose a Gaussian latent-error assumption and is therefore reported as a robustness check alongside the Tobit model when normality is doubtful. The optimization status for this fit is converged after 108 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units; the reported coefficients are back-transformed to the original judgement scale. Standard errors come from the ctqr asymptotic covariance matrix and should be interpreted as complementary robustness evidence rather than clustered Tobit replacements.

9.4 H3: Interaction Moderation

This section tests the interaction and moderation between absolute empathy dimensions and the outgroup status of the perpetrator.

9.4.1 Model A: Composite Empathy Interaction

Table 16: H3 Model A: Composite Empathy Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-9.592	6.858	0.162
Empathy composite (average)	-1.668	2.337	0.475
Outgroup perpetrator (ref = ingroup)	-2.077	2.725	0.446
Control label hidden (ref = ingroup)	-2.658	3.346	0.427
Victim outgroup (ref = ingroup)	0.278	0.910	0.760
Observer role (ref = victim)	0.545	0.798	0.494
Engineering participant (ref = humanities)	0.188	1.556	0.904
Man (ref = woman)	-1.765	1.744	0.312
Age	0.478	0.241	0.047
Socioeconomic status	-0.247	0.706	0.726
Negotiator 2 (ref = negotiator 1)	0.124	0.458	0.787
Empathy x outgroup perpetrator	0.885	1.245	0.477
Empathy x control label hidden	1.578	1.589	0.321
Scale variance (log)	1.993	0.096	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -9.592 (p=0.162). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -1.668. The p-value of 0.475 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -2.077. The p-value of 0.446 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of -2.658. The p-value of 0.427 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.278. The p-value of 0.760 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.545. The p-value of 0.494 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.188. The p-value of 0.904 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -1.765. The p-value of 0.312 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.478. The p-value of 0.047 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.247. The p-value of 0.726 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:perp_outgroup (representing the combined interaction effect between overall empathy and an outgroup perpetrator) has an estimated β coefficient of 0.885. The p-value of 0.477 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:perp_control (representing the combined interaction effect between overall empathy and an unidentified perpetrator) has an estimated β coefficient of 1.578. The p-value of 0.321 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.993, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.934, p = 8.055e-18). Consequently, standard clustered errors might be inefficient. For maximum robustness in future analyses, we recommend exploring a non-parametric alternative such as Censored Least Absolute Deviations (CLAD) or specifying bootstrapped standard errors for the Tobit estimator.

Table 17: H3 Model A: Composite Empathy Regression Coefficients (CLAD robustness).

label	estimate	std_error	p_value
Intercept	-12.967	7.538	0.085
Empathy composite (average)	-1.611	1.551	0.299
Outgroup perpetrator (ref = ingroup)	-1.443	3.948	0.715
Control label hidden (ref = ingroup)	-4.722	5.435	0.385
Victim outgroup (ref = ingroup)	-0.328	0.532	0.537
Observer role (ref = victim)	1.093	0.669	0.102
Engineering participant (ref = humanities)	0.066	0.496	0.894
Man (ref = woman)	-0.803	0.706	0.256
Age	0.667	0.286	0.020
Socioeconomic status	-0.551	0.304	0.070
Negotiator 2 (ref = negotiator 1)	0.093	0.429	0.828
Empathy x outgroup perpetrator	0.624	1.891	0.741
Empathy x control label hidden	2.443	2.704	0.366

CLAD Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -12.967 (p=0.085). The term iri_total (representing the average composite of empathetic propensity across the participant) has an estimated β coefficient of -1.611. The p-value of 0.299 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -1.443. The p-value of 0.715 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of -4.722. The p-value of 0.385 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of -0.328. The p-value of 0.537 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 1.093. The p-value of 0.102 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.066. The p-value of 0.894 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.803. The p-value of 0.256 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's

biological age in years) has an estimated β coefficient of 0.667. The p-value of 0.020 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.551. The p-value of 0.070 indicates the effect is marginally significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:perp_outgroup (representing the combined interaction effect between overall empathy and an outgroup perpetrator) has an estimated β coefficient of 0.624. The p-value of 0.741 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_total:perp_control (representing the combined interaction effect between overall empathy and an unidentified perpetrator) has an estimated β coefficient of 2.443. The p-value of 0.366 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The CLAD robustness model was estimated as an interval-censored median regression (p = 0.50). This estimator does not impose a Gaussian latent-error assumption and is therefore reported as a robustness check alongside the Tobit model when normality is doubtful. The optimization status for this fit is not confirmed as converged after 2000 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units; the reported coefficients are back-transformed to the original judgement scale. Standard errors come from the ctqr asymptotic covariance matrix and should be interpreted as complementary robustness evidence rather than clustered Tobit replacements.

9.4.2 Model B: Separated Constructs Interaction

Table 18: H3 Model B: Separated Constructs Regression Coefficients (Tobit).

label	estimate	std_error	p_value
Intercept	-11.669	7.497	0.120
Empathy: Fantasy scale	-1.347	1.766	0.446
Outgroup perpetrator (ref = ingroup)	-0.751	3.182	0.813
Empathy: Empathic concern	-1.294	2.639	0.624
Empathy: Perspective taking	2.558	1.815	0.159
Empathy: Personal distress	-1.397	1.866	0.454
Control label hidden (ref = ingroup)	-1.828	4.018	0.649
Victim outgroup (ref = ingroup)	0.180	0.880	0.838
Observer role (ref = victim)	0.425	0.783	0.587
Engineering participant (ref = humanities)	0.530	1.660	0.749
Man (ref = woman)	-1.794	1.825	0.325
Age	0.489	0.233	0.036
Socioeconomic status	-0.300	0.688	0.663
Negotiator 2 (ref = negotiator 1)	0.140	0.452	0.757
Fantasy x outgroup perpetrator	-0.571	1.157	0.622
Empathic concern x outgroup perpetrator	-0.959	1.289	0.457
Perspective taking x outgroup perpetrator	-0.953	1.145	0.405
Personal distress x outgroup perpetrator	3.287	1.317	0.013
Fantasy x control label hidden	-0.107	1.477	0.942
Empathic concern x control label hidden	0.102	1.568	0.948
Perspective taking x control label hidden	0.357	1.836	0.846
Personal distress x control label hidden	0.840	1.247	0.501
Scale variance (log)	1.976	0.093	0.000

Tobit Estimator Interpretation

The Intercept represents the baseline latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, the baseline is estimated at -11.669 (p=0.120). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -1.347. The p-value of 0.446 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -0.751. The p-value of 0.813 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated

β coefficient of -1.294. The p-value of 0.624 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 2.558. The p-value of 0.159 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -1.397. The p-value of 0.454 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of -1.828. The p-value of 0.649 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of 0.180. The p-value of 0.838 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.425. The p-value of 0.587 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.530. The p-value of 0.749 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -1.794. The p-value of 0.325 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.489. The p-value of 0.036 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.300. The p-value of 0.663 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:perp_outgroup (representing the combined interaction effect between the Fantasy scale and an outgroup perpetrator) has an estimated β coefficient of -0.571. The p-value of 0.622 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_ec (representing the combined interaction effect between Empathic Concern and an outgroup perpetrator) has an estimated β coefficient of -0.959. The p-value of 0.457 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_pt (representing the combined interaction effect between Perspective Taking and an outgroup perpetrator) has an estimated β coefficient of -0.953. The p-value of 0.405 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_pd (representing the combined interaction effect between Personal Distress and an outgroup perpetrator) has an estimated β coefficient of 3.287. The p-value of 0.013 indicates statistically significant, meaning the predictor has a positive effect on the latent moral judgment. The term iri_fs:perp_control (representing the combined interaction effect between the Fantasy scale and an unidentified perpetrator) has an estimated β coefficient of -0.107. The p-value of 0.942 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_control (repre-

senting the combined interaction effect between Empathic Concern and an unidentified perpetrator) has an estimated β coefficient of 0.102. The p-value of 0.948 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:perp_control (representing the combined interaction effect between Perspective Taking and an unidentified perpetrator) has an estimated β coefficient of 0.357. The p-value of 0.846 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_control (representing the combined interaction effect between Personal Distress and an unidentified perpetrator) has an estimated β coefficient of 0.840. The p-value of 0.501 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The Log(scale) is a standard Tobit variance parameter representing the natural logarithm of the standard deviation of the unobserved residuals. The estimated scale parameter log is 1.976, capturing the underlying dispersion of latent judgments.

Diagnostics

The Shapiro-Wilk test on the deviance residuals indicates a violation of the normality assumption (W = 0.951, p = 3.347e-15). Consequently, standard clustered errors might be inefficient. For maximum robustness in future analyses, we recommend exploring a non-parametric alternative such as Censored Least Absolute Deviations (CLAD) or specifying bootstrapped standard errors for the Tobit estimator.

Table 19: H3 Model B: Separated Constructs Regression Coefficients (CLAD robustness).

label	estimate	std_error	p_value
Intercept	-10.262	19.150	0.592
Empathy: Fantasy scale	-0.364	12.100	0.976
Outgroup perpetrator (ref = ingroup)	-2.753	18.819	0.884
Empathy: Empathic concern	-0.558	12.911	0.966
Empathy: Perspective taking	0.920	3.652	0.801
Empathy: Personal distress	-1.927	4.670	0.680
Control label hidden (ref = ingroup)	-6.331	24.214	0.794
Victim outgroup (ref = ingroup)	-0.630	1.079	0.559
Observer role (ref = victim)	0.974	0.699	0.164
Engineering participant (ref = humanities)	0.136	0.651	0.835
Man (ref = woman)	-0.961	3.102	0.757
Age	0.528	0.488	0.280
Socioeconomic status	-0.551	0.572	0.335
Negotiator 2 (ref = negotiator 1)	-0.052	0.518	0.920
Fantasy x outgroup perpetrator	-0.592	11.581	0.959
Empathic concern x outgroup perpetrator	0.385	11.557	0.973
Perspective taking x outgroup perpetrator	-0.941	5.443	0.863
Personal distress x outgroup perpetrator	2.562	3.742	0.494
Fantasy x control label hidden	-0.569	11.930	0.962
Empathic concern x control label hidden	0.961	15.695	0.951
Perspective taking x control label hidden	1.385	4.819	0.774
Personal distress x control label hidden	1.052	4.292	0.806

CLAD Robustness Estimator Interpretation

The Intercept represents the baseline conditional median latent moral judgment when all continuous predictors are zero and categorical predictors are at their reference levels. In this model, that baseline median is estimated at -10.262 (p=0.592). The term iri_fs (representing the participant's inclination to transpose themselves imaginatively into the feelings of fictitious characters (Fantasy scale)) has an estimated β coefficient of -0.364. The p-value of 0.976 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup (representing whether the perpetrator belonged to a faculty different from the participant (Outgroup)) has an estimated β coefficient of -2.753. The p-value of 0.884 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec (representing the participant's tendency to experience feelings of sympathy and compassion for unfortunate others (Empathic Concern)) has an estimated β coefficient of -0.558. The p-value of 0.966 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt (representing the participant's tendency to spontaneously adopt the psychological point of view of others (Perspective Taking)) has an estimated β coefficient of 0.920. The p-value of

0.801 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd (representing the participant's tendency to experience distress and discomfort in tense interpersonal settings (Personal Distress)) has an estimated β coefficient of -1.927. The p-value of 0.680 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_control (representing whether the perpetrator's organizational alignment was explicitly hidden (Control label)) has an estimated β coefficient of -6.331. The p-value of 0.794 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term victim_outgroup (representing whether the victim was affiliated with a different faculty than the participant) has an estimated β coefficient of -0.630. The p-value of 0.559 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term role_observer (representing the procedural role where the participant acted exclusively as an observer rather than a victim) has an estimated β coefficient of 0.974. The p-value of 0.164 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term participant_engineering (representing whether the participant belonged to the Engineering faculty as opposed to Humanities) has an estimated β coefficient of 0.136. The p-value of 0.835 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term sex_man (representing whether the participant identified as a man instead of a woman) has an estimated β coefficient of -0.961. The p-value of 0.757 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term age (representing the participant's biological age in years) has an estimated β coefficient of 0.528. The p-value of 0.280 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term economic_status (representing the socioeconomic contextual stratum of the participant's background) has an estimated β coefficient of -0.551. The p-value of 0.335 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:perp_outgroup (representing the combined interaction effect between the Fantasy scale and an outgroup perpetrator) has an estimated β coefficient of -0.592. The p-value of 0.959 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_ec (representing the combined interaction effect between Empathic Concern and an outgroup perpetrator) has an estimated β coefficient of 0.385. The p-value of 0.973 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_pt (representing the combined interaction effect between Perspective Taking and an outgroup perpetrator) has an estimated β coefficient of -0.941. The p-value of 0.863 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term perp_outgroup:iri_pd (representing the combined interaction effect between Personal Distress and an outgroup perpetrator) has an estimated β coefficient of 2.562. The p-value of 0.494 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_fs:perp_control (representing the combined interaction effect between the Fantasy scale and an unidentified perpetrator) has an estimated β coefficient of -0.569. The p-value of 0.962 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_ec:perp_control (representing the combined interaction effect between Empathic Concern and an unidentified perpetrator) has an estimated β coefficient of 0.961. The p-value of 0.951 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pt:perp_control

(representing the combined interaction effect between Perspective Taking and an unidentified perpetrator) has an estimated β coefficient of 1.385. The p-value of 0.774 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector. The term iri_pd:perp_control (representing the combined interaction effect between Personal Distress and an unidentified perpetrator) has an estimated β coefficient of 1.052. The p-value of 0.806 indicates the effect is not statistically significant, meaning we do not have enough evidence to reject the null hypothesis for this vector.

Diagnostics

The CLAD robustness model was estimated as an interval-censored median regression (p = 0.50). This estimator does not impose a Gaussian latent-error assumption and is therefore reported as a robustness check alongside the Tobit model when normality is doubtful. The optimization status for this fit is converged after 108 iterations. To satisfy the positive-time requirement of the censored quantile routine, the bounded judgement outcome was shifted internally by 10.0 units; the reported coefficients are back-transformed to the original judgement scale. Standard errors come from the ctqr asymptotic covariance matrix and should be interpreted as complementary robustness evidence rather than clustered Tobit replacements.

10 Discussion and Limitations

While the dual-estimator strategy strengthens the analysis, several limitations remain. First, the Tobit clustering logic assumes independence between participants, which may be violated by shared institutional affiliations. Second, we assume a normal distribution for the latent errors ϵ_{ij} in the Tobit branch; departures from normality could affect the consistency of the maximum likelihood estimators. The CLAD robustness branch reduces dependence on that assumption, but its current implementation relies on asymptotic covariance estimates rather than clustered sandwich errors. Finally, the use of unstandardized averages for empathy assumes a linear mapping between the psychometric scale and the latent moral preference across both estimators.

11 Conclusion

Based on the combined interval-censored Tobit estimations and CLAD robustness checks, the structural effects of empathy, group dynamics, and betrayal have been documented alongside their theoretical assumptions above.